

LEVEL-BY-LEVEL RATIONALITY AND THE SHAPOVALOV SUMMABILITY PROBLEM FOR THE VIRASORO SPECTRAL R -MATRIX

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ABSTRACT. Level-by-level rationality of the bar-cobar spectral R -matrix for Vir_c is proved (bar-cobar A_∞ operations plus Shapovalov inversion: poles at the Kac determinant zero locus). Convergence of the level sum on the principal series requires a summable level majorant. Fixed-basis condition-number control for the Shapovalov matrices is one input, but a polynomial lower bound cannot by itself control exponential numerators and partition growth. The Kac determinant supplies a product formula and finite-level nonvanishing away from the null-vector locus; it does not supply a least-eigenvalue lower bound. Numerical data through level 8 are a fixed-PBW-basis conditioning check. The full identification with the Ponsot–Teschner fusion kernel also requires the four-point channel, contour, normalization, and external Liouville momenta to be matched. The missing inputs are the summable majorant and the channel comparison. Relative to the five-family theorem of Vol. I [1], the Virasoro spectral R -matrix is the adjacent class- $\mathcal{M}$ analogue of the Shapovalov-witness principle carried by reflection-equation Shapovalov on types B/C/D; the rank-1 Baxter hyperplane is a type- $\mathcal{A}$ rational artifact (Vol. I, Proposition “Baxter hyperplane as a type- $\mathcal{A}$ rational artifact”).

I. STATEMENT AND REDUCTION

Theorem 1.1 (Conditional identification with PT). *Fix the real Liouville chamber $b \in \mathbb{R}_{>0}$, $b^2 \notin \mathbb{Q}$, $Q = b + b^{-1}$, and $c = 1 + 6Q^2$. Fix a Ponsot–Teschner four-point block [7, 9] with external momenta $\alpha_i = Q/2 + iP_i$, internal s - and t -channel momenta α_s, α_t , compact real principal-series ranges for the $P_\bullet$, the standard contour separating the two PT pole families, and the PT normalization of conformal blocks. After complexifying the $P_\bullet$ -variables, remove the singular hyperplanes $P_\bullet = \pm i(rb + sb^{-1})/2$ and keep the contour chamber fixed. Assume Conjecture 2.1 below on this compact chamber and assume that the bar-cobar spectral R -matrix is expressed in this same channel and normalization and satisfies the full b, b^{-1} shift system there. Then the bar-cobar spectral R -matrix of Vir_c equals the Ponsot–Teschner fusion kernel on that principal-series chamber. Along any complex b -path on which both sides are analytically continued with the same pole prescription, contour chamber, normalization, and full shift system, this equality continues by the same uniqueness theorem.*

The conditional identification has one fixed-level algebraic input and two proof obligations: a summable level majorant and the normalized PT channel comparison.

Proposition 1.2 (Level-by-level rationality). *At fixed levels (N_1, N_2) , each matrix element of the bar-cobar spectral $R^{\text{bar}, (N_1, N_2)}$ is a rational function of the central charge and the weights $h_i = \alpha_i(Q - \alpha_i)$, with possible poles only along the finite-level Kac determinant divisors $\det G_{N_j}(c, h_i) = 0$; equivalently, in Liouville coordinates, along $h_i = h_{r,s}(c)$ for $rs \leq N_j$.*

Proof. A_∞ operations m_k are polynomial in spectral parameters and central charge; evaluation on the Verma basis is polynomial in h_i ; homotopy transfer introduces Gram (Shapovalov) inverses, contributing poles at the Kac determinant zeros $h = h_{r,s}$. At each finite level the number of operations is finite, and rationality is preserved. $\square$

Proposition 1.3 (Eberhardt uniqueness, normalized full-shift form). *A meromorphic function of the Liouville momenta satisfying the full b and b^{-1} shift system, the Ponsot–Teschner symmetry and normalisation conditions, and the principal-series analyticity hypotheses is unique up to the stated normalisation. Under that normalisation it is the Ponsot–Teschner kernel [2, Theorem 1.1].*

For the bar-cobar spectral R -matrix, one must verify the full shift system, symmetry, and normalisation before invoking Proposition 1.3. The analytic problem is convergence of the level sum $R^{\text{bar}} = \sum_{N=0}^{\infty} R^{(N)}$ on compact subsets of the principal-series domain.

Proof of Theorem 1.1. Proposition 1.2 places every fixed-level term on the complement of the finite Kac divisor in the chosen Liouville chamber. Conjecture 2.1 supplies a Weierstrass majorant on compact subsets of the chamber complement, so the level sum defines a meromorphic kernel there with the stated pole prescription. The channel, contour, external-momentum, and normalization hypotheses identify this kernel with a candidate in the same four-point problem as the PT kernel. The full b , b^{-1} shift system, symmetry, and analyticity hypotheses are exactly the inputs of Proposition 1.3; hence the two kernels coincide on the chamber. Analytic continuation along a path preserving the same pole prescription and contour chamber carries equality of the continued solutions. $\square$

2. THE CONVERGENCE QUESTION

Conjecture 2.1 (Summable level bound). *Fix $b \in \mathbb{R}_{>0}$ with $b^2 \notin \mathbb{Q}$ and a compact subset of the real unitary principal-series chamber $h = Q^2/4 + P^2$. In the complexified P -plane, remove the Kac hyperplanes $P = \pm i(rb + sb^{-1})/2$ and work on a compact subset of one chamber of the complement. There the level- N contribution to the bar-cobar spectral R -matrix satisfies*

$$\|R^{(N)}\| \leq M_N, \quad \sum_{N \geq 0} M_N < \infty.$$

Equivalently, the Shapovalov denominator estimates, numerator cancellations, and partition growth combine to give a Weierstrass majorant.

At $c = 25$, $h = 2$ ($b = 1$, $P = 1$) the fixed partition/PBW-basis conditioning computation gives:

N	1	2	3	4	5	6	7	8
$\lambda_{\min}$	4	14.8	49.2	118.5	245.4	408.9	668.8	975.7
$\lambda_{\min}/p(N)$	4.0	7.4	16.4	23.7	35.1	37.2	44.6	44.4

This data point lies at rational $b^2 = 1$, outside the irrational b^2 hypothesis above. It is a numerical check on Shapovalov conditioning, not evidence for the summable majorant by itself.

3. ANALYTIC PROOF OBLIGATIONS FOR THEOREM 1.1

3.0.0.1. Denominator. On the real unitary principal series, after fixing conventions for $Q = b + b^{-1}$, $h = Q^2/4 + P^2$, and $P_{r,s} = (rb + sb^{-1})/2$,

$$h - h_{r,s} = P^2 + P_{r,s}^2.$$

The Kac determinant gives the finite-level product formula [3, 6]

$$\det G_N = C_N \prod_{rs \leq N} (P^2 + P_{r,s}^2)^{p(N-rs)}$$

in the chosen PBW normalization, with $C_N \neq 0$. This proves finite-level nonvanishing away from the Kac locus, not Conjecture 2.1. Determinants do not control least singular values without condition-number information. The expected route is a direct condition-number estimate for G_N , using the block decomposition of descendants into L_{-1} -chains and interlacing under partition refinement.

3.o.o.2. Numerator. The level- N contribution to the R -matrix involves iterated $\mathcal{A}_\infty$ wheel integrals. The presently available estimate is only the naive Wick/tree count: the $k(\mathbb{C})$ volume times the OPE coefficient bound, with no cancellation. This gives no better than a $C^N \cdot N!$ envelope for N mode contractions, hence no summable majorant.

3.o.o.3. Tree refinement. The crude $N!$ is too weak, and even a polynomial lower bound on $\lambda_{\min}(G_N)$ cannot control an exponential numerator after partition growth is included. The needed estimate is the summable bound of Conjecture 2.1: either the numerator has subexponential decay after $\mathcal{A}_\infty$ cancellations, or the Shapovalov inverse has a stronger effective denominator than the present determinant estimate gives.

3.o.o.4. Roussillon route. Roussillon [8] gives a convergent series representation of F^{PT} in the normalization and parameter regime treated there. To use it here one must first match the PT four-point channel, contour, normalization, and external momenta, and then match the bar-cobar series term by term. Only under that additional comparison does meromorphicity follow from Roussillon's convergence analysis.

4. POSITION INSIDE THE MC₃ FIVE-FAMILY MECHANISM

The five-family thick-generation theorem of Vol. I [1] supplies the Shapovalov-witness template. Its five unconditional evaluation-generated rows are:

Family	Shapovalov witness	Thick-generation status
Type $\mathcal{A}$ rational Yangian	asymptotic prefundamentals (Hernandez–Jimbo)	unconditional
Types B/C/D twisted Yangian	reflection-equation Shapovalov ($\text{sdet } K(u)$)	unconditional
Uniform simple $\mathfrak{g}$	Chari–Moura multiplicity-free ℓ -weights	unconditional
Elliptic $\mathfrak{sl}_N$	theta-divisor complement (Felder–Varchenko)	unconditional (theta-divisor)
Super $\mathfrak{gl}_{m n}$	parity-balance hyperplane (Nazarov Berezinian)	unconditional (hyperplane)
Virasoro bar-cobar	summable Shapovalov/level majorant on principal series	conjectural

The Virasoro line is not a sixth case of the five-family theorem. It is the class- $\mathcal{M}$ analogue whose comparison theorem is Theorem 1.1 and whose missing analytic input is Conjecture 2.1. The five-family mechanism retires the rank-1 Baxter hyperplane as a *type- $\mathcal{A}$ rational artifact* (Vol. I, Proposition “Baxter hyperplane as a type- $\mathcal{A}$ rational artifact”): the Baxter constraint $b = a - \frac{1}{2}$ is visible only in the rational-form presentation of $Y(\mathfrak{sl}_N)$ and does not persist under degeneration from any other chart. In the present Virasoro setting the analogous codimension-1 locus is the Kac determinant zero locus $h = h_{r,s}$, which is globally intrinsic (not chart-dependent); the corresponding Shapovalov witness is the fixed-basis conditioning datum, while the summable level majorant is still unproved.

Remark 4.1 (Silent non-coverage). The MC₃ full-DK-generation theorem of Vol. I is silent on the following families outside the evaluation-generated core: logarithmic $\mathcal{W}(p)$ -algebras (finite-dimensional Zhu algebra does not imply a uniform Massey bound; Gurarie and Flohr exhibit logarithmic amplitudes obstructing that route [4, 5], and tempering itself is open), $N = 2$ superconformal algebras, cosets (beyond irrational cosets), non-rational lattice VOAs, and roots of unity. The conditional Theorem 1.1 applies only to the chosen irrational principal-series Virasoro chamber, which excludes the logarithmic $\mathcal{W}(p)$ class entirely.

5. ANALYTIC PROOF OBLIGATIONS

- (i) Extend the fixed-PBW Shapovalov conditioning computation to $N = 12$ – 15 with arbitrary-precision arithmetic and test several principal-series points, including irrational b^2 .

- (ii) Prove condition-number estimates for G_N via the block structure of Virasoro descendants into L_{-1} -chains and interlacing under partition refinement. The Kac determinant itself supplies only product/nonvanishing data.
- (iii) Replace the crude $N!$ numerator bound by the Stasheff-polytope bound, using the $\mathcal{A}_\infty$ relations to organise the cancellations.
- (iv) Prove a summable majorant for the complete level contribution, including partition growth. Only then apply the Weierstrass M-test on compact subsets of the principal series.
- (v) Fix the PT four-point channel, contour, external momenta, and normalization; verify the normalized full b, b^{-1} shift system, symmetry, and normalisation hypotheses of Proposition 1.3; then conclude $R^{\text{bar}} = F^{\text{PT}}$.

6. DEPENDENCIES

The argument depends on and feeds:

- Vol. I, “MC₃ on the evaluation-generated core, five-family mechanism”;
- Vol. I, “Baxter hyperplane as a type- $\mathcal{A}$ rational artifact”;
- Vol. I, “Full DK_g generation status” for the scope beyond the evaluation-generated core;
- Vol. I, “Chiral bialgebra equivalence on the Koszul locus”, which carries the ordered bar-cobar R -matrix equivalence;
- Eberhardt difference-Galois uniqueness [2, Theorem 1.1].

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